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DIGITAL FORENSICS INVESTIGATION REPORT

Case Study: Ransomware Attack at Sama Solutions Sdn. Bhd.

1.0 Introduction

Digital forensics is the process of identifying, collecting, examining, analyzing, and presenting digital evidence to investigate incidents involving computers and information systems. It is important in cybersecurity because it helps organizations understand how security incidents happen, what systems are affected, and how similar incidents can be prevented.

According to Kent et al. (2006), digital forensic techniques support computer security investigations by helping investigators collect and analyze information from files, operating systems, applications, and network activities. Proper evidence handling is necessary to maintain the integrity and reliability of investigation findings.

Ransomware is a type of malicious software that can prevent users from accessing their files or systems and demand payment to restore access. Ransomware incidents can interrupt business operations, affect important information, and cause financial losses. The Cybersecurity and Infrastructure Security Agency (CISA, 2023) recommends that organizations prepare appropriate prevention, detection, response, and recovery procedures to reduce these risks.

This report presents a simulated ransomware investigation at Sama Solutions Sdn. Bhd., a fictional IT services company in Kuala Lumpur, Malaysia.

Our five-member investigation team was assigned to examine the suspected ransomware incident, identify relevant digital evidence, reconstruct the sequence of events, and recommend security improvements.

The report describes the incident background, investigation objectives, forensic procedures, evidence collection, analysis, findings, and recommendations.

2.0 Incident Background and Description

2.1 Company Background

Sama Solutions Sdn. Bhd. is a fictional IT services company located in Kuala Lumpur, Malaysia. The company has approximately 80 employees and provides software development, technical support, and IT consulting services.

The company uses computers and a shared file server to store and manage important business information, including customer records, employee information, financial reports, and project documents.

The Finance Department regularly communicates with external suppliers and receives invoices through email. This makes email an important communication method for daily financial activities.

2.2 Incident Description

On 12 October 2026, at approximately 8:42 AM, a Finance Department employee received an unexpected email claiming to be from a company supplier.

The email contained the subject "October Invoice – Payment Confirmation Required" and referred to an attachment named Invoice_Oct2026.zip.

At approximately 8:46 AM, the employee downloaded the attachment. According to the simulated employee statement, the employee opened the file at 8:49 AM to review the invoice.

Shortly afterward, the workstation identified as FIN-PC-07 began showing unusual activity.

At 8:51 AM, the endpoint security system recorded a suspicious process alert. At approximately 8:54 AM, several financial documents became inaccessible, and their filenames were changed to include the .locked extension.

At 8:56 AM, a file named README_RECOVER.txt appeared on the workstation. The message claimed that company files had been encrypted and demanded cryptocurrency payment to restore access.

The employee reported the incident to the IT Department at 9:02 AM.

At 9:08 AM, the IT Department disconnected FIN-PC-07 from the company network to reduce the possibility of further damage.

The company then established a five-member digital forensic investigation team to examine the incident.

2.3 Possible Business Impact

The suspected ransomware incident created several risks for Sama Solutions.

First, the Finance Department could not access some important financial documents. This could delay financial reporting, supplier payments, and other daily business activities.

Second, unusual file activity was recorded in the shared Finance folder on the company's file server, FS-01. This created concerns about possible damage to shared company information.

Third, the incident could affect the confidentiality, integrity, and availability of company information. However, our simulated evidence does not confirm that customer information was stolen or that the attack spread to other computers.

Therefore, these risks are considered potential impacts rather than confirmed losses.

3.0 Investigation Objectives and Scope

3.1 Investigation Objectives

The main objective of our investigation was to identify the nature and possible cause of the ransomware incident at Sama Solutions.

The specific objectives were:

1. To identify the affected computer and possible affected network resources.
2. To investigate the suspicious email and downloaded attachment.
3. To identify and preserve relevant digital evidence.
4. To analyze workstation and network activity records.
5. To reconstruct the incident timeline.
6. To determine the possible impact of the ransomware incident.
7. To recommend containment, recovery, and preventive security measures.

3.2 Investigation Scope

The investigation focused on the Finance Department workstation FIN-PC-07 and its recorded interaction with file server FS-01.

The simulated evidence included the suspicious email, workstation logs, affected-file examples, ransom note, and network access logs.

The investigation did not include the actual execution of ransomware, examination of a real malicious program, or a full forensic examination of a production company network.

All activities were presented as a fictional investigation based on prepared evidence records.

4.0 Investigation Team and Responsibilities

The investigation team consisted of five members with different responsibilities.

Table 1: Digital Forensic Investigation Team

Member	Investigation Role	Main Responsibility
Member 1	Incident Response Team Leader	Coordinate the investigation and manage the incident response
Member 2	Digital Evidence Collection Officer	Identify, collect, preserve, and document evidence
Member 3	Digital Forensic Analyst	Examine evidence and reconstruct the incident timeline

Member 4	Cybersecurity Incident Response Specialist	Assess containment, recovery, and security improvements
Member 5	Investigation Documentation Officer	Prepare investigation findings, forms, and the final report

Each member contributed to a different stage of the simulated investigation.
The team leader coordinated the activities, while the other investigators handled evidence documentation, analysis, incident response, and reporting.
This division of responsibilities helped our group demonstrate how a professional digital forensic investigation team could operate.

5.0 Digital Forensic Investigation Methodology

Our investigation followed the general forensic process described by Kent et al. (2006), which includes collection, examination, analysis, and reporting.
We also included incident identification, evidence preservation, containment, and recovery planning.
Nelson et al. (2025) explain that incident response should be integrated into an organization's cybersecurity risk management activities to improve detection, response, and recovery.

5.1 Incident Identification

The investigation began after the Finance Department employee reported inaccessible documents and the appearance of a ransom note.
We reviewed the fictional incident report and identified FIN-PC-07 as the primary affected workstation.
We also identified file server FS-01 as a possible affected resource because the workstation accessed a shared Finance folder during the incident.
Based on the initial information, we classified the incident as suspected ransomware activity.

5.2 Evidence Preservation and Collection

After identifying the incident, our team established an evidence collection plan.
We identified five important digital evidence items associated with the incident.
In the simulated investigation, the Evidence Collection Officer recorded the evidence descriptions, sources, collection times, and responsible officers.
The team assumed that authorized personnel created forensic copies of the relevant data and stored them in a protected evidence location.
SHA-256 hashing was included in the preservation procedure to describe how investigators could verify

the integrity of digital files. However, no actual hashes were calculated during this classroom simulation. A separate Evidence Collection Form and Chain of Custody Document were prepared to record the fictional handling procedures.

5.3 Evidence Examination

The Digital Forensic Analyst examined the prepared evidence records to identify suspicious activities.
The email record was reviewed to identify the suspicious sender information, subject, and attachment reference.
The workstation logs were examined to identify the attachment download, suspicious process alert, file modifications, and isolation event.
The affected-file records were reviewed to identify changes to document filenames.
The ransom note was examined to understand the attacker's displayed demand.
Finally, the network logs were reviewed to identify unusual access to the Finance shared folder.

5.4 Evidence Analysis

During the analysis stage, we compared the information from different evidence items.
The timeline indicated that the suspicious email was received before unusual workstation activity began.
The recorded events also showed that several filenames changed shortly before the appearance of the ransom note.
These observations were consistent with suspected ransomware activity.
However, we could not conclusively prove that the email attachment contained ransomware because the original malicious file contents were not available.
The exact ransomware family and complete infection mechanism also remained unknown.

5.5 Reporting

After completing the simulated analysis, our team documented the incident timeline, evidence observations, investigation findings, limitations, and recommended security measures.
The findings were organized into this report to help the company's management understand the incident and make informed security decisions.

6.0 Evidence Collection and Preservation

Digital evidence must be documented carefully to support an investigation. Kent et al. (2006) emphasize the importance of preserving evidence integrity and maintaining reliable documentation during forensic activities.

For this simulated incident, our team identified five evidence items.

Table 2: Digital Evidence Collection Summary

Evidence ID	Evidence Description	Simulated Collection Time
EV-001	Suspicious invoice email	9:30 AM
EV-002	Workstation activity logs	9:35 AM
EV-003	Affected-file examples	9:40 AM
EV-004	Ransom note	9:45 AM
EV-005	Network and file access logs	9:50 AM

6.1 EV-001: Suspicious Email

The first evidence item was the suspicious invoice email received by the Finance employee.
The email contained an invoice-related subject and referenced a ZIP attachment.
This evidence was important because it helped investigators identify a possible entry point for the suspected attack.
However, the available email record did not conclusively prove that the attachment contained malicious software.

6.2 EV-002: Workstation Activity Logs

The second evidence item consisted of simulated activity records from FIN-PC-07.
The logs showed an attachment download, a suspicious process alert, file modifications, and the workstation isolation event.
These records helped our team understand the sequence of activities leading to the incident.

6.3 EV-003: Affected-File Examples

The third evidence item represented three affected Finance documents:

- Budget2026.xlsx.locked
- Payroll_Oct2026.xlsx.locked
- SupplierPayments.pdf.locked

The filenames indicated that several documents had been changed during the incident.
However, the filename extensions alone were not sufficient to prove that the actual file contents had been encrypted.

6.4 EV-004: Ransom Note

The fourth evidence item was a fictional ransom note named README_RECOVER.txt.
The note stated that company files had been encrypted and requested cryptocurrency payment in exchange for restoring access.
This evidence supported the classification of the incident as suspected ransomware activity.

6.5 EV-005: Network and File Access Logs

The fifth evidence item contained simulated network records.
The logs indicated that FIN-PC-07 accessed the shared Finance folder on FS-01 at 8:52 AM.
Unusual file-write activity was recorded at approximately 8:55 AM.
This suggested possible impact on the shared folder. However, the available information did not confirm that the server was infected or that other computers were compromised.

6.6 Chain of Custody

Chain of custody refers to the documentation of how evidence is collected, handled, transferred, and stored.
In our fictional investigation, Member 2 acted as the Evidence Collection Officer and recorded the collection details.
Member 3 received verified working copies for forensic analysis, while Member 5 received the investigation findings for documentation.
The original evidence was assumed to remain in protected storage under the responsibility of the evidence custodian.
A separate Chain of Custody Document records the simulated handling and transfer of evidence.

7.0 Forensic Examination and Analysis

7.1 Investigation Timeline

Our team developed an incident timeline using the fictional email, workstation, and network records.

Table 3: Ransomware Incident Timeline

Time	Incident Activity
8:42 AM	Suspicious invoice email received
8:46 AM	ZIP attachment downloaded
8:49 AM	Employee reportedly opened the attachment

8:51 AM	Suspicious process activity detected
8:52 AM	Finance shared folder accessed
8:54 AM	Files began changing to .locked
8:55 AM	Unusual shared-folder write activity recorded
8:56 AM	Ransom note appeared
9:02 AM	Employee reported incident
9:08 AM	Workstation isolated
9:20 AM	Initial network review completed
9:30 AM	Simulated evidence collection began
10:30 AM	Simulated forensic analysis began
2:00 PM	Preliminary findings presented

7.2 Analysis of the Initial Attack

The investigation showed that suspicious activity began shortly after the Finance employee downloaded and reportedly opened the invoice attachment.
The email and download records suggested that the incident might have started through a phishing-related attachment.
However, the original executable file was not available for verification.
Therefore, we identified the email attachment as a suspected initial attack vector rather than a confirmed infection source.

7.3 Analysis of Affected Files

The workstation logs recorded changes to several Finance documents at approximately 8:54 AM.
The affected documents had filenames ending with .locked.
These changes were followed by the appearance of a ransom note at 8:56 AM.
The relationship between these events strongly suggested ransomware-related file activity.
However, the investigation did not independently verify the encryption of the original document contents.

7.4 Analysis of Network Activity

The network evidence showed that FIN-PC-07 accessed the Finance shared folder on FS-01 before the ransom note appeared.
Unusual file-write activity was also recorded on the shared folder.
This raised concerns about possible damage to files stored on the company server.

However, the available logs did not provide enough evidence to confirm server compromise or additional infected computers.

The initial network review at 9:20 AM found no other confirmed affected endpoints within the available records.

7.5 Investigation Limitations

Our investigation had several limitations.

First, the incident was fictional, and no real ransomware sample was examined.

Second, the original malicious attachment contents were unavailable.

Third, the exact ransomware family could not be determined.

Fourth, the investigation could not independently confirm encryption of the affected files.

Finally, the evidence did not establish whether confidential information had been stolen or transferred outside the company.

These limitations were considered when preparing our final conclusions.

8.0 Investigation Findings

Based on the five simulated evidence items, our team identified the following findings.

Finding 1: Suspicious Email as a Possible Entry Point

The Finance employee received an unexpected invoice email shortly before the incident. This email was considered a possible initial attack vector, although its malicious nature was not conclusively proven.

Finding 2: Abnormal Workstation Activity

Workstation FIN-PC-07 recorded a suspicious process alert and unusual file modifications after the employee reportedly opened the downloaded attachment.

Finding 3: Ransomware-Related File Changes

Several financial documents were renamed with the .locked extension. A ransom note appeared shortly afterward, indicating activity consistent with ransomware.

Finding 4: Possible Shared-Folder Impact

Unusual write activity was recorded on the shared Finance folder on FS-01. However, the available evidence did not confirm that the server was infected.

Finding 5: Initial Containment Completed

The IT Department disconnected the affected workstation from the network and restricted access to relevant resources. These actions helped reduce the possibility of further damage during the initial response.

Overall, the investigation concluded that the available evidence was consistent with a suspected

ransomware incident affecting FIN-PC-07. However, the exact infection method and full impact could not be confirmed.

9.0 Incident Response and Security Recommendations

9.1 Incident Containment and Recovery

After the incident was reported, the IT Department isolated FIN-PC-07 from the network.
The team also restricted write access to the Finance shared folder and temporarily limited access to the affected employee account while verification was pending.
The simulated response team preserved relevant records and reviewed available network information.
The company's backup records were scheduled for verification before any recovery activities.
CISA (2023) recommends isolating affected systems, preserving forensic evidence, examining the possible scope of compromise, and restoring systems from trusted backups after appropriate security checks.
For Sama Solutions, we recommend that the affected workstation should remain isolated until the security team confirms that malicious activity has been removed.
File restoration should only begin after backups have been checked and the recovery environment is considered safe.

9.2 Security Recommendations

Based on our findings, we recommend the following improvements.

1. Improve Email Security

Sama Solutions should strengthen email filtering to identify suspicious attachments and messages. Employees should also receive awareness training to recognize phishing attempts.

2. Implement Stronger Endpoint Protection

The company should use updated endpoint security tools to identify suspicious processes, unusual file modifications, and possible ransomware behavior.

3. Improve Backup and Recovery Procedures

Important business information should be backed up regularly. The company should maintain protected offline or immutable backups and test recovery procedures.

4. Strengthen Access Control

Employees should only receive the system and file permissions required for their responsibilities. This can reduce the possible damage caused by compromised accounts.

5. Apply Network Segmentation

The company's network should be divided into appropriate security zones to reduce unnecessary communication between departments and critical servers.

6. Conduct Incident Response Training

The company should provide cybersecurity awareness training and conduct regular incident response exercises.

Nelson et al. (2025) explain that effective incident response requires planning and integration with organizational cybersecurity risk management.

These recommendations can help Sama Solutions improve its ability to prevent, detect, respond to, and recover from future security incidents.

10.0 Conclusion

This report presented a fictional digital forensic investigation of a suspected ransomware incident at Sama Solutions Sdn. Bhd. on 12 October 2026.

Our five-member investigation team examined five simulated evidence items, including a suspicious email, workstation activity logs, affected-file examples, a ransom note, and network access logs.

The investigation found that unusual workstation activity began shortly after a Finance Department employee received and reportedly opened a suspicious invoice attachment.

Several financial documents were renamed with the .locked extension, and a ransom note appeared on the affected workstation.

The network records also indicated unusual activity involving the company's shared Finance folder. However, the investigation could not conclusively determine whether the server was compromised or whether confidential information was stolen.

The IT Department isolated the affected workstation and initiated protective measures to reduce further risks.

Based on our investigation, we recommended improvements in email security, endpoint protection, backup procedures, access control, network segmentation, and employee awareness.

Through this simulated investigation, our group demonstrated the importance of following proper forensic procedures, preserving digital evidence, maintaining chain of custody documentation, and preparing accurate investigation findings.

The case also showed that a successful incident response requires cooperation between investigators, IT personnel, and company management.

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